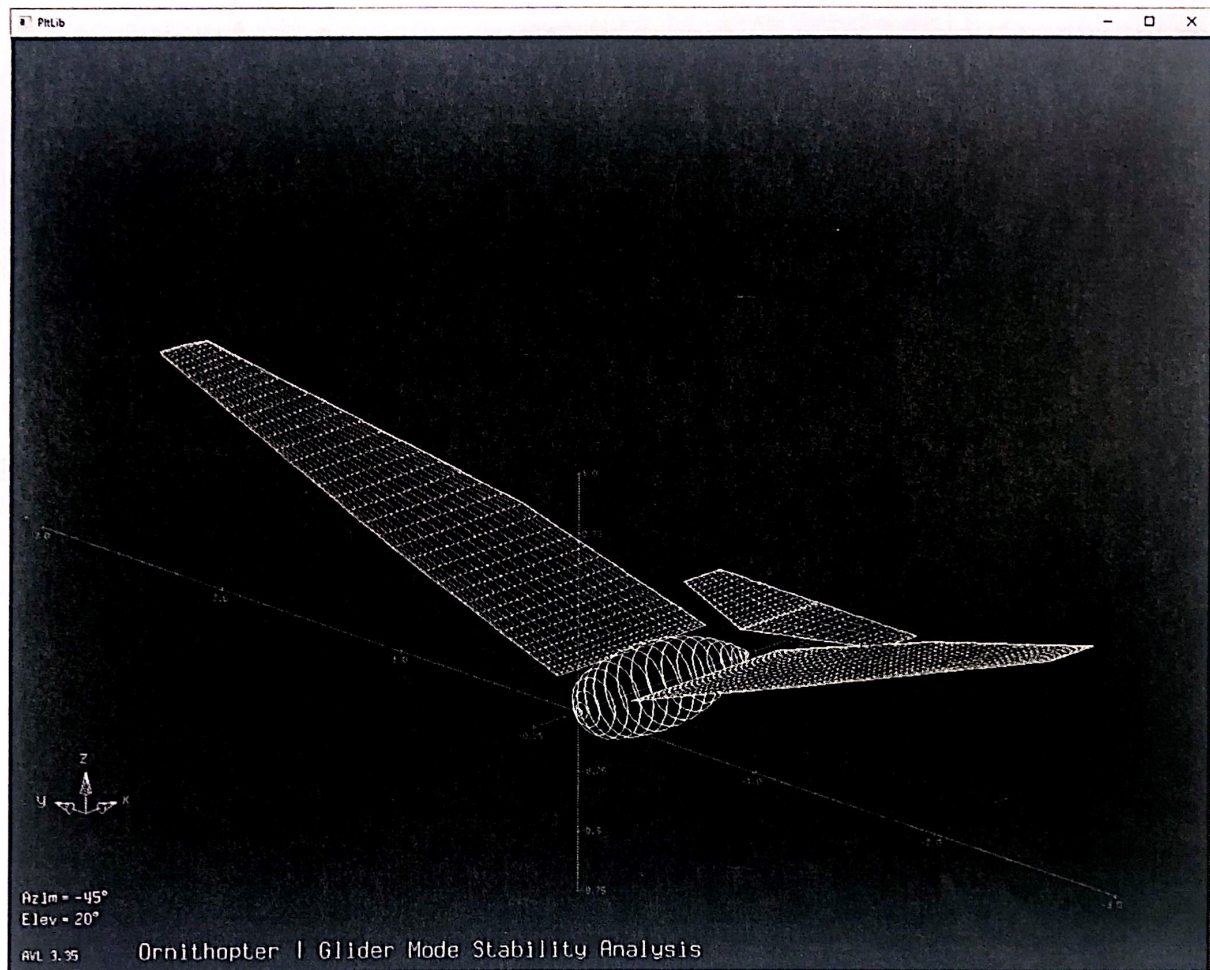
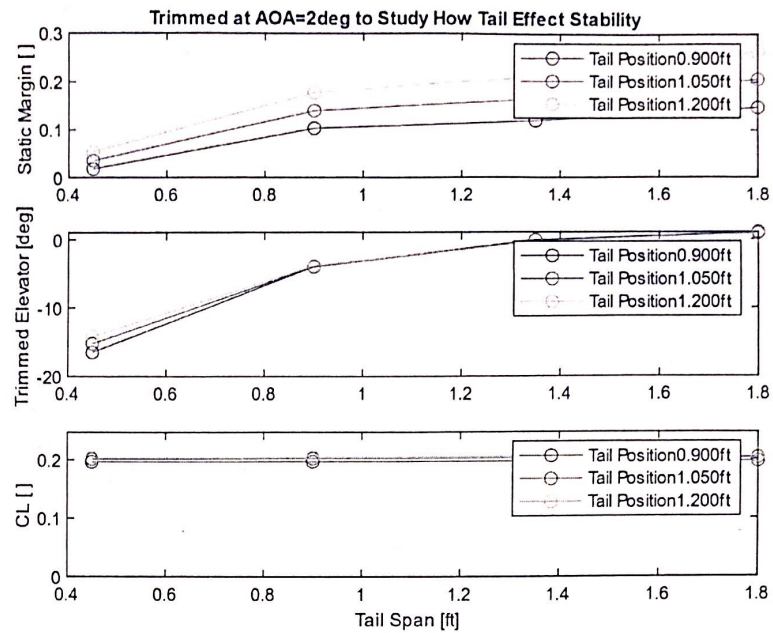


### Question 1

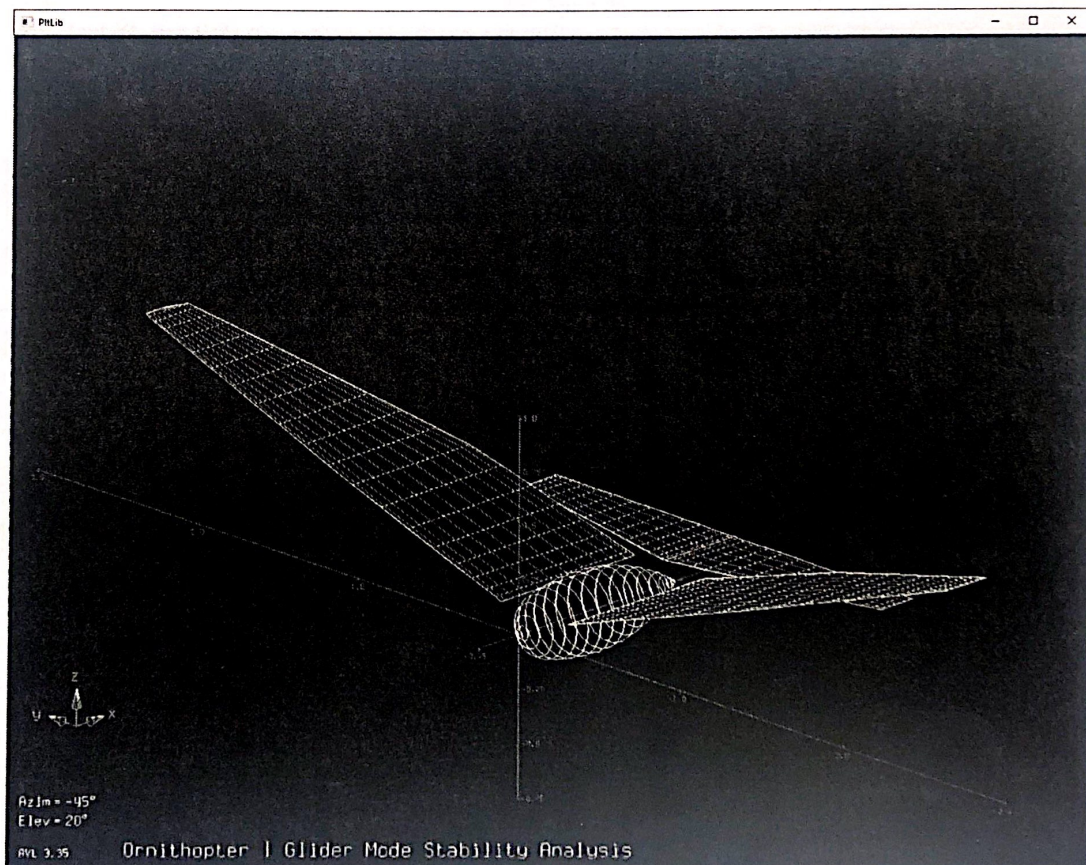
Originally, the ornithopter looks like this:



Design sweep of AVL (vertices-lattice method) suggested that I go for a large tail span to reach high static margin for glider mode

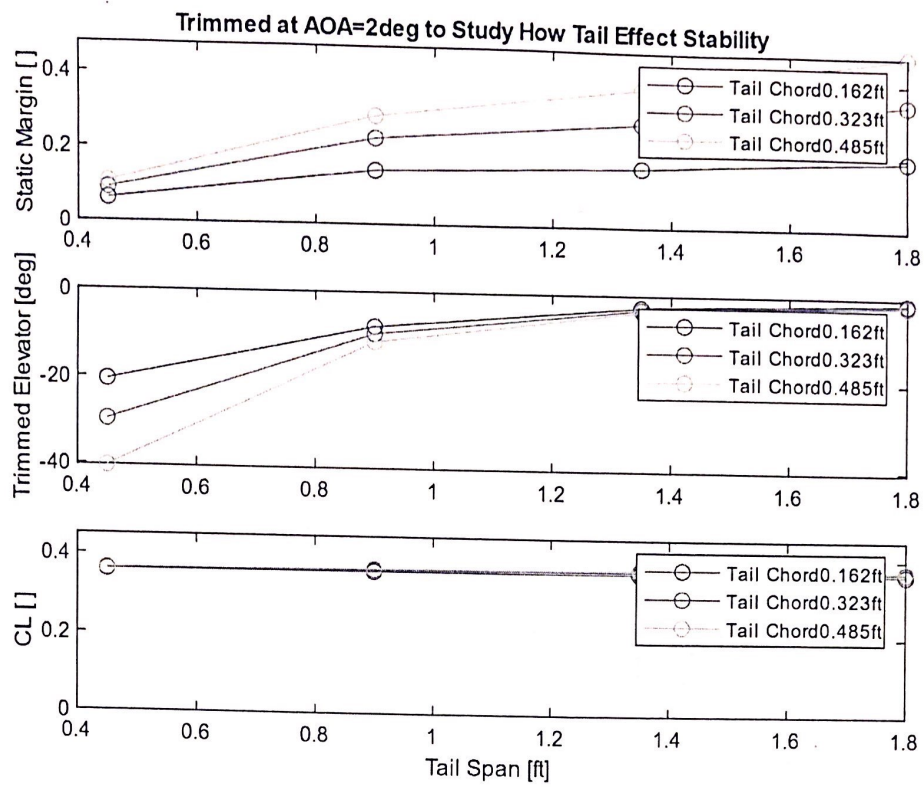


However, the suggested 1.8ft tail span is too large in terms of fabrication.

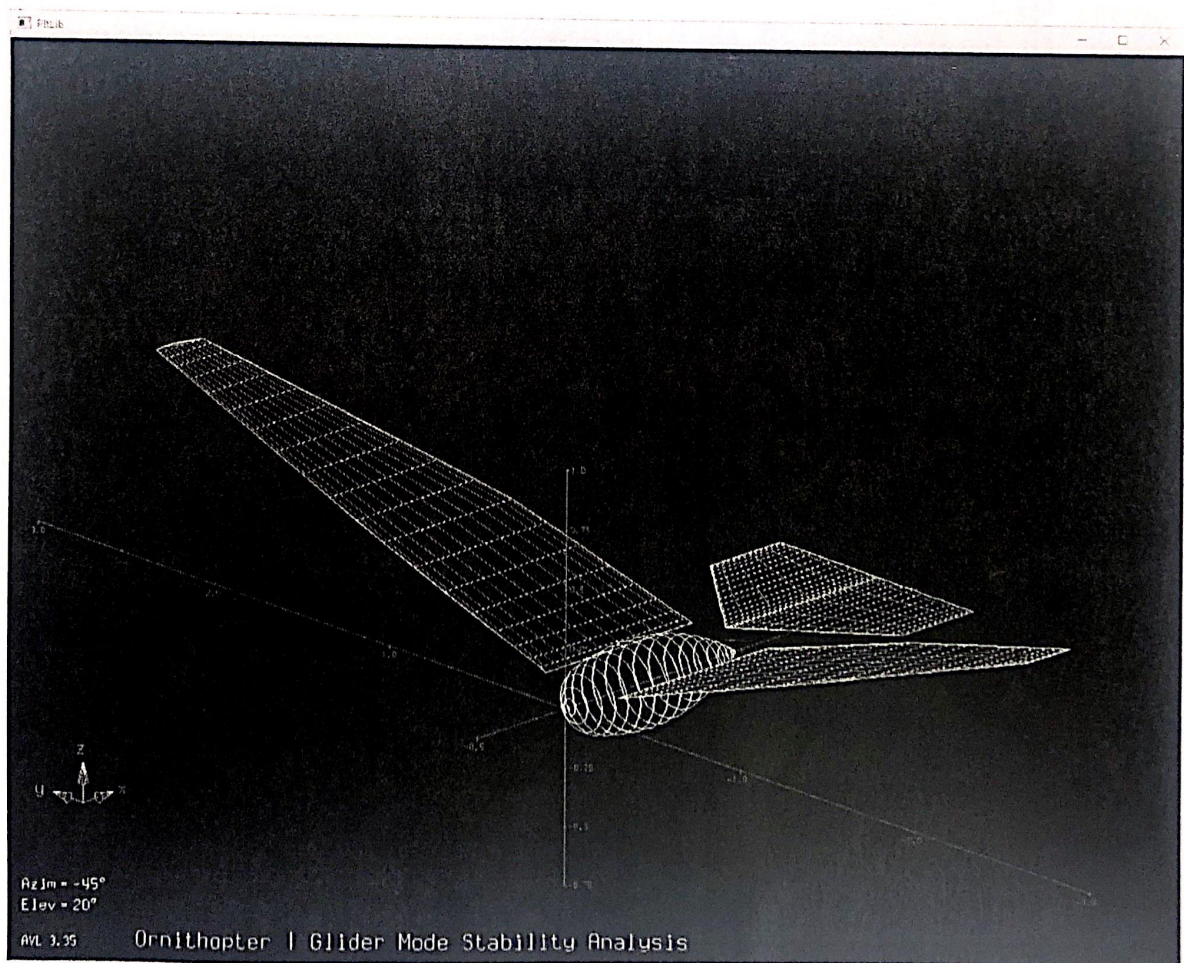


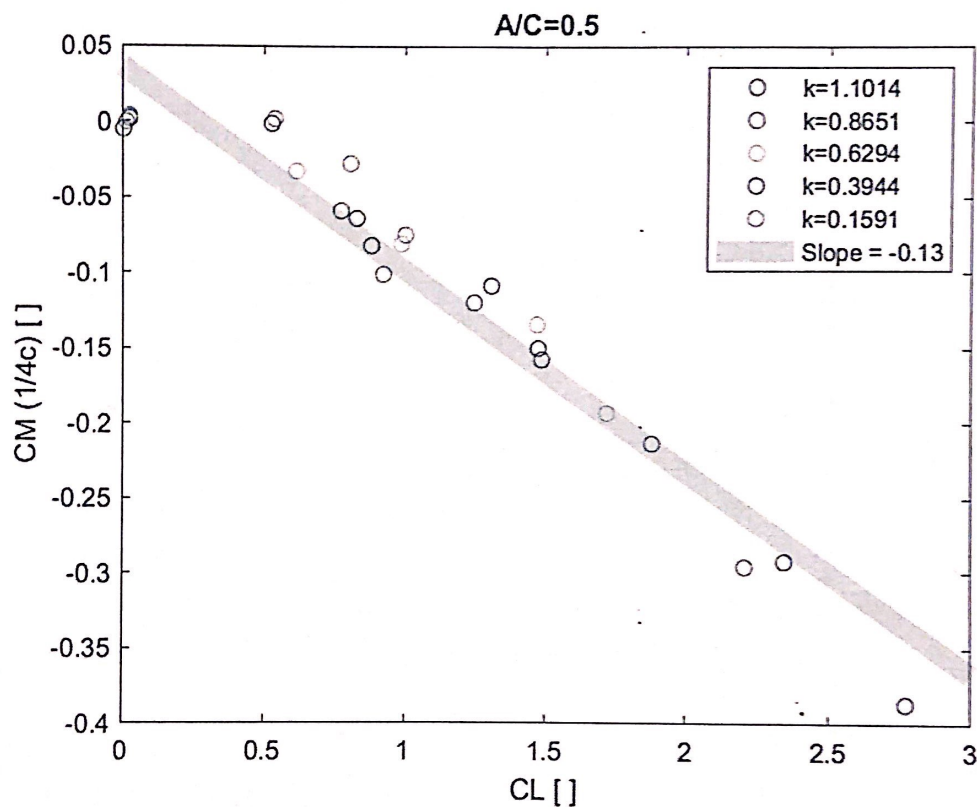
Further design sweep suggests that, instead of increasing the tail span, I should consider increasing tail chord.





Now the tail looks like this





Data from Chierighin, N. et al. "Unsteady Lift and Moment of a Periodically Plunging Airfoil". AIAA Journal, vol. 57, no. 1, American Institute of Aeronautics and Astronautics. 2019, pp. 208-22, doi: 10.2514/1.5057639